

Journal 04: An Overview of Artificial Intelligence Ethics

Sharie Rhea

Southern New Hampshire University

CS 530: Artificial Intelligence Principles and Applications

Joshua Andrews

March 1, 2026

Artificial intelligence (AI) ethics has become increasingly significant as AI systems shape individual experiences, social institutions, and broader environmental contexts. This paper outlines key ethical principles and implementation approaches, evaluates technological solutions for transparency and related concerns, discusses challenges in applying ethical theories to AI systems, and assesses the importance of international guidelines and future ethical frameworks. Together, these perspectives illustrate both the promise and the complexity of building ethical AI systems.

There are different ways to apply ethics to AI. One method is virtue ethics, in which virtues such as transparency, fairness, accountability, ownership, dignity, and security guide decisions (Huang et al., 2022; Liu et al., 2025). Another approach is to follow a known set of moral rules, otherwise referred to as deontology (Huang et al., 2022). Deontology is a natural fit for AI systems, as computers excel at following predefined rulesets. Utilitarianism offers a third framework, focusing on maximizing benefits and minimizing harm at either the individual level or across broader populations (Huang et al., 2022). Regardless of which theoretical lens is chosen, ethical decision-making in AI can be implemented through either top-down or bottom-up approaches, each with its own merits. Top-down systems encode predefined rules, while bottom-up models learn from environmental input and data patterns (Huang et al., 2022).

Despite these structured approaches, ethical AI remains deeply challenging in practice. The principal reason creating an unbiased and fully ethical AI system is nearly impossible is that such systems rely on formal rationality (Nishant et al., 2023). Humans, while biased, can shift between ethical theories, weigh competing values, and adapt decisions to the unique context of a situation. Patterns, biases, or rules that work in one scenario do not necessarily apply in another (Nishant et al., 2023). As situations grow more complex and nuanced, AI performance often

deteriorates (Nishant et al., 2023). In this sense, purely technological solutions to ethical AI are insufficient. Human oversight, auditing, and validation remain the most reliable methods for ensuring responsible outcomes (R & C, 2025).

Beyond technical limitations, applying ethical theories to AI raises broader societal concerns. Although AI provides significant benefits, it is necessary to consider whether those benefits outweigh the potential consequences. Unregulated AI can contribute to privacy erosion, propaganda dissemination, job displacement, discrimination, commercialization of data, hyper-personalization, and heightened societal fear and mistrust (Liu et al., 2025; Yugendran et al., 2025). For example, modern surveillance systems rely on AI for facial recognition and analysis (R & C, 2025). While such systems may enhance public safety, they also generate serious concerns regarding privacy rights, explicit consent, and discriminatory outcomes (R & C, 2025). Another concern is the increasing polarization of social media which may lead to feedback loops as AI generates more and more divisive content (Nishant et al., 2023). However, at the same time, there is a risk of overcorrection. Efforts to “overclean” AI systems by eliminating all bias may unintentionally introduce opposing distortions. Humans naturally rely on pattern recognition shaped by repeated experiences, and attempting to remove every trace of bias may create new imbalances instead (Nishant et al., 2023).

Given these complexities, the development of international regulations and ethical guidelines is increasingly urgent. One useful method categorizes AI’s impacts into individual, societal, and environmental effects (Huang et al., 2022). However, establishing and enforcing global standards presents its own challenges. Cultural differences shape perspectives on ethical behavior, and individual judgments vary widely. Moreover, responsibility for ethical AI is distributed across multiple stakeholders. Users must engage with AI responsibly, companies

must prioritize user well-being over purely commercial interests, and developers must ensure appropriate training data, rigorous testing, and ongoing auditing (Huang et al., 2022; Liu et al., 2025). The IEEE Code of Ethics also provides a strong foundation for developers to apply ethical principles consistently across technological domains, not just in AI (IEEE, 2020). Finally, even if proper standards are established, enforcing them presents its own set of challenges in opaque systems.

In conclusion, AI ethics demands both philosophical rigor and practical accountability. While ethical theories offer valuable guidance, their implementation in AI systems remains constrained by technological, social, and cultural limitations. As AI continues to expand its influence, sustained human oversight, collaborative governance, and adaptable ethical frameworks will be essential to ensuring that these systems serve the broader good rather than undermine it.

References

- Huang, C., Zhang, Z., Mao, B., & Yao, X. (2022). An Overview of Artificial Intelligence Ethics. *IEEE Transactions on Artificial Intelligence*, 4(4), 1–21.
<https://doi.org/10.1109/tai.2022.3194503>
- IEEE. (2020). IEEE Code of Ethics | IEEE. [iee.org](https://www.ieee.org/about/corporate/governance/p7-8).
<https://www.ieee.org/about/corporate/governance/p7-8>
- Liu, A. C.-C., Law, O. M. K., & Law, I. (2025). *Machine Learning, Data Science Foundations with Python*. zyBook. <https://learn.zybooks.com/zybook/CS-530-10427.202617-1>
- Nishant, R., Schneckenberg, D., & Ravishankar, M. N. (2023). The formal rationality of artificial intelligence-based algorithms and the problem of bias. *Journal of Information Technology*, 39(1). <https://doi.org/10.1177/02683962231176842>
- R, L., & C, S. (2025). Social and Ethical Challenges of Artificial Intelligence in Surveillance and Security Systems. 2025 9th International Conference on Computational System and Information Technology for Sustainable Solutions (CSITSS), 1–6.
<https://doi.org/10.1109/csitss67709.2025.11294538>
- Yugendran V., Jenefa, L., Jayakumar, M., & M. Sakthivel. (2025). Ethics of AI Technology Interruption and Discomfort on Employee Turnover. 1–6.
<https://doi.org/10.1109/icdsaai65575.2025.11011775>